



Dublin and Dún Laoghaire ETB

Programme Module for

Web Development

leading to

Level 6 QQI

Web Development 6N1277

Introduction

This programme module may be delivered as a standalone module leading to certification in a QQI minor award. It may also be delivered as part of an overall validated programme leading to a Level 6 QQI Certificate.

The teacher/tutor should familiarise themselves with the information contained in Dublin and Dún Laoghaire ETB's programme descriptor for the relevant validated programme prior to delivering this programme module.

The programme module is structured as follows:

1. Title of Programme Module Web Development
2. QQI Component Title and Code Web Development 6N17881
3. Duration in hours 150
4. Credit Value of QQI Component 15 Credits
5. Status
6. Special Requirements
7. Aim of the Programme Module The programme module aims to provide the learner with knowledge, skills and capacity to design, develop and update modern, dynamic and responsive websites.
8. Objectives of the Programme Module
9. Learning Outcomes
10. Indicative Content
11. Assessment a. Assessment Technique(s) b. Mapping of Learning Outcomes to Assessment Technique(s) c. Guidelines for Assessment Activities
12. Grading
13. Learner Marking Sheet(s), including Assessment Criteria

Integrated Delivery and Assessment

The teacher/tutor is encouraged to integrate the delivery of content where an overlap between content of this programme module and one or more other programme modules is identified. This programme module will facilitate the learner to develop the academic and vocational language, literacy and numeracy skills relevant to the themes and content of the module.

Likewise the teacher/tutor is encouraged to integrate assessment where there is an opportunity to facilitate a learner to produce one piece of assessment evidence which demonstrates the learning outcomes from more than one programme module. The integration of the delivery and assessment of level 6 Communications and level 6 Mathematics modules with that of other level 6 modules is specifically encouraged, as appropriate.

Indicative Content

The indicative content in Section 10 does not cover all teaching possibilities. The teacher/tutor is encouraged to be creative in devising and implementing other approaches, as appropriate. The use of examples is there to provide suggestions. The teacher/tutor is free to use other examples, as appropriate. The indicative content ensures all learning outcomes are addressed but it may not follow the same sequence as that in which the learning outcomes are listed in Section 9. It is the teacher's/tutor's responsibility to ensure that all learning outcomes are included in the delivery of this programme module.

1. Title of Programme Module Web Development
2. Component Name and Code Web Development 6N1277
3. Duration in Hours 150 Hours (typical learner effort, to include both directed and self directed learning)
4. Credit Value 15 Credits
5. Status This programme module may be compulsory or optional within the context of the validated programme. Please refer to the relevant programme descriptor, Section 9 Programme Structure
6. Special Requirements NONE
7. Aim of the Programme Module This programme module aims to enable students to develop and assess modern, standards compliant websites, using appropriate tools.
8. Objectives of the Programme Module • Understand the history and development of the World Wide Web up to the current day • Design and develop standards compliant websites, using HTML and CSS • Use scripting languages to, securely, develop and maintain multi-tier dynamic websites • Analyse and assess websites with regard to usability, accessibility and visibility to search engines. • To assist the learner to develop the academic and vocational language, literacy and numeracy skills related to Web Development through the medium of the indicative content • To enable the learner to exercise substantial personal autonomy and responsibility.

9. Learning Outcomes of Level 6 Web Development 6N1277

Learners will be able to:

1. Determine how and when to use Cascading Style Sheets (CSS) and demonstrate understanding of the differences between CSS versions and their uses
2. Describe a multi-tier web architecture to support dynamic web content
3. Outline the differences between Web 1.0, Web 2.0 and beyond
4. Explain and illustrate the difference between responsive and non-responsive web design
5. Examine the development and evolution of the most recent version of HTML and create web pages with the most recent HTML compliant tags
6. Use at least one industry standard tool or CMS to develop websites.
7. Address security issues associated with implementing dynamic web content
8. Perform usability and accessibility assessments on new and existing websites
9. Write, implement, and maintain web scripts using a secure scripting language
10. Develop web pages and content that conform to international standards e.g. W3C, Universal Design, and Accessibility
11. Develop and test cross platform / cross browser websites
12. Create an interactive website integrating a wide range of multimedia content
13. Discuss the format of web pages and their contents including html code up to latest versions
14. Employ dynamic and persistent data technologies e.g. JSON
15. Implement web design techniques such as wire-framing for conceptual layouts
16. Host, maintain and update a variety of website content
17. Utilise search engine optimisation (SEO) techniques to successfully promote a website.

10. Indicative Content
Section 1 : Html&CSS History and Development
Outcomes 3, 4, 5, 13 Outline the difference between Web 1.0 , Web 2.0 and beyond. Facilitate the learner to • Investigate the history of HTML to include Web 1.0, Web 2.0 and beyond Explain and illustrate the Difference between responsive and non - responsive web design Facilitate the learner to understand: • The term Responsive Web Design • Describe the difference between Responsive, Adaptive and Mobile websites Enable the learner to • Create pages using flexible layouts • Create pages using media queries • Use flexible media in web page creation Examine the development and evolution of the most recent version of HTML and Create web pages with the most recent HTML compliant tags Discuss the format of web pages and their contents including html code up to latest versions Enable the Learners to : • Define the term HTML • Be aware of different versions of HTML • Identify deprecated and non-deprecated tags and attributes • Illustrate a working knowledge of tags and attributes Learners should know how to: • Migrate from one version of HTML to another • Understand HTML Browser Support

Learners should be able to

- Add New Elements to HTML.
- Understand What Semantic Elements are?,
- New Semantic Elements in HTML to include :<article>,<aside>, <details>, <figcaption>, <figure>, <footer>, <header>, <main>, <mark> <nav> <section> <summary> <time>
- Be able to validate their HTML markup using HTML validators

Section 2: Design and develop standards compliant websites

Outcomes 1, 8, 10, 11, 12 15,

Determine how and when to use Cascading Style Sheets (CSS) and demonstrate an understanding of the differences between CSS versions and their uses

Facilitate the learner to understand

- A Definition of CSS and understanding of CSS CSS3
- The benefits of CSS
- CSS specifications WC3
- CSS rules (selector, property, value, style inheritance, specificity)
- Default styles
- Browser support for CSS

Develop and test cross platform / cross browser websites

Perform usability and accessibility assessments on new and existing websites

- Facilitate the user to assess a new or existing website with regard to:
 - Site Navigation
 - Readability
 - Site Speed
 - User Experience
- Examine the accessibility of a site through:
 - Use of HTML accessibility tags
 - Semantic HTML markup
 - Colour Choice

- Cross Browser/Platform compatibility

Develop web pages and content that conform to international standards e.g. W3C, Universal Design, and Accessibility

Create an interactive website integrating a wide range of multimedia content

Facilitate learners to understand the importance of

- Structure: layout of pages; navigation; format of content and cascading style sheets (CSS); interactive features e.g. catalogue of products, shopping cart; images; animation
- Content: proofed, correct and appropriate; information source; structured for purpose e.g. prose, bullets, tables
- Tools and techniques: navigation diagram e.g. linear, hierarchy, matrix; building interactivity tools e.g. animation; audio/visual elements; ensuring compliance with W3C; meta tagging; cascading style sheets
- Review: functionality testing e.g. user environments, links, navigation; content; check against user requirements; user acceptance; audit trail of changes.

Implement web design techniques such as wire-framing for conceptual layouts

Facilitate learners to understand how to:

Evaluate Why Wireframes are used

Understand the purpose of wire-framing and use of design techniques in the following design areas:

- **Layout:** What interface elements will be used? Where will the elements be placed? Which items are related to each other and should be grouped?
- **Priority:** What are the key elements that you want users to notice first? What do you want a user to do when he or she arrives at a page?
- **Content:** What content should be present on a page? Is text enough or should there be images and videos as well? Which options should UI elements like drop-down lists or pop-up menus contain?
- **Navigation:** How will users move around and interact with the site? Will there be a site-wide navigation menu or any other guidance? Does the website provide a search function?
- **Functionality and behaviour:** How does a page respond to user input? Will an action

require a full page refresh? How will form validation errors be handled?

- Be aware of various Wireframing tools e.g. Powermockup, Balsamiq Mockup
- Use wireframes or similar design techniques for website design
- Appropriate use of colors in relation to color theory

Section 3 A Hosting a Website

Learning Outcomes 16, 17

Host, maintain and update a variety of website content

Facilitate the learner to understand the following:

Hosting a Site

- Introduction to servers
- Understand the different types of hosting providers
- Choosing server features
- Getting a domain name
- Purchasing a domain

Setting up a Coding Environment

- Setting up FTP accounts
- Using FTP client
- Granting additional FTP access

Setting up Emails

- Setting up webmail and forwards

Password Protection

- Protecting user information
- Protecting directories

Update website content

- Keep website content current

Utilise search engine optimisation (SEO) techniques to successfully promote a website.

Facilitate learners to understand:

- What is Search Engine Optimization (SEO)?
- Why does my website need SEO?
- How search engines operate
- How people interact with search engines
- Keyword Research
- The impact of usability and user experience on search engine ranking
- The importance of Links in SEO
- Search engine tools and services
- Web analytics

Section 4 Using scripting Languages

Outcomes 2, 6, 7, 9, 14

Describe a multi-tier web architecture to support dynamic web content

- Explain the terms client, server, middleware
- Explain the functions of database server and explain how it differs from a desktop database system.
- Describe the benefits of a multi-tier architecture
- Provide an Overview and describe the benefits of Multitier Architecture

Use industry standard tools or CMS to develop websites.

- Explain the functions of content management system.
- Describe where a CMS might be more appropriate for a client.
- Download and install a CMS, e.g. WordPress and install it on a local web server for development and testing.
- Use themes to customise the appearance of the site

- Customise elements of themes to make it individual to the site. e.g. Change images, layout, etc.
- Create custom formats for user content. E.g. custom posts in Wordpress.
- Create Menus to organise the content of the site logically.

Address security issues associated with Dynamic Web Content.

- Describe common attacks on dynamic web sites, e.g. SQL injection, cross site scripting
- Test all SQL for vulnerability to SQL injection
- Use a vulnerability scanner to test websites for vulnerabilities, e.g. skipfish
- Use correct encryption techniques for the saving of sensitive data in databases

Write implement and maintain web scripts using a secure scripting language

Server Side Scripting

- Create a script to connect to a database server and retrieve some data
- Write code that demonstrates common ways of displaying results from a database query, e.g. tables, lists
- Create scripts that allow the creation, updating and deletion of data in the database. Using HTML forms and appropriate server side scripts.

Client Side Scripting

- Use client side scripts to:
 - Validate form input
 - Manipulate content on a page for effect, e.g. sliding panels, cycling images, etc.
 - Use the Document Object Model to get and set attributes of elements on a page, e.g.
 - Textboxes
 - labels
 - Do basic calculations, e.g. calculate the totals on an order page

- Get and set cookie values

- Use Ajax to perform asynchronous updates to a page

Employ Dynamic and Persistent Data Technologies

- Create a script to parse and display data from a JSON file.
- Use HTML5 persistent data

11. Assessment**11a. Assessment Techniques****Skills Demonstration 60% (Practical)****Exam (Theory) 40%****11b. Mapping of Learning Outcomes to Assessment Techniques**

In order to ensure that the learner is facilitated to demonstrate the achievement of all learning outcomes from the component specification; each learning outcome is mapped to an assessment technique(s). This mapping should not restrict an assessor from taking an integrated approach to assessment.

Learning Outcome	Assessment Technique
1. Determine how and when to use Cascading Style Sheets (CSS) and demonstrate understanding of the differences between CSS versions and their uses	Skills Demonstration
2. Describe a multi-tier web architecture to support dynamic web content	Exam
3. Outline the differences between Web 1.0, Web 2.0 and beyond	Exam
4. Explain and illustrate the difference between responsive and non-responsive web design	Exam
5. Examine the development and evolution of the most recent version of HTML and create web pages with the most recent HTML compliant tags	Skills Demonstration & Exam
6. Use at least one industry standard tool or CMS to develop websites.	Skills Demonstration
7. Address security issues associated with implementing dynamic web content	Exam
8. Perform usability and accessibility assessments on new and existing websites	Skills demonstration
9. Write, implement, and maintain web scripts using a secure scripting language	Skills Demonstration
10. Develop web pages and content that conform to international standards e.g. W3C, Universal Design, and Accessibility	Skills Demonstration
11. Develop and test cross platform / cross browser websites	Skills Demonstration
12. Create an interactive website integrating a wide range of multimedia content	Skills Demonstration

13.	Discuss the format of web pages and their contents including html code up to latest versions	Exam
14.	Employ dynamic and persistent data technologies e.g. JSON	Skills Demonstration
15.	Implement web design techniques such as wire-framing for conceptual layouts	Skills Demonstration
16.	Host, maintain and update a variety of website content	Skills Demonstration
17.	Utilise search engine optimisation (SEO) techniques to successfully promote a website	Skills Demonstration

11c. Guidelines for Assessment Activities

The assessor is required to devise assessment briefs and marking schemes for the Skills Demonstrations/examination papers, marking schemes and outline solutions for the examination. In devising the assessment briefs/examination papers care should be taken to ensure that the learner is given the opportunity to show evidence of achievement of ALL the learning outcomes. Assessment briefs may be designed to allow the learner to make use of a wide range of media in presenting assessment evidence, as appropriate. Quality assured procedures must be in place to ensure the reliability of learner evidence.

Exam	40%
Learner will undertake a written examination of not less than two hours duration.	
The learner will complete a written exam containing 5 structured questions covering the following Learning Outcomes 2, 3, 4, 5, 7, 13 The exam will consist of 5 structured questions: 8 marks each Structured questions are divided into a number of related parts and generally require the learner to demonstrate more in-depth knowledge and understanding of a topic. All instructions for the learner must be clearly outlined in an examination paper	

Skills Demonstration 1	Assessment Technique 2 30%
The learner will complete an assignment over a period of no more than 4 – 6 weeks	
Skills Demonstration 1 35%	
Over multiple sessions throughout the year as the required learning outcomes have been delivered, the learner will demonstrate skills from learning outcomes 1, 5, 6, 8, 9 10, 11, 12 14, 15, 16. 17	
The assessor will devise a brief which will require students to develop a website. The website should cover the following areas:	
SD 1 a Design a website LO 15 Use appropriate wire-framing techniques to design a website appropriate to its purpose.	
SD 1b Create a website LOs 1, 5, 17, 10 ,12 - All HTML validates correctly via W3C HTML Validation Service - Appropriate use of CSS selectors to target specific elements - Appropriate use of relevant CSS properties to style page elements as specified in the brief - Appropriate use of colors based on an understanding of color theory - Appropriate use of size units for modifying element properties such as margin, padding, border size - Demonstrated understanding of typographical concepts. - Demonstrated understanding of accessibility concepts and best practices. - Demonstrated understanding of current HTML and its uses. - Demonstrated understanding of current CSS and its uses. - Appropriate use of absolute and relative URLs for related images, CSS files, and hyperlinks. - All links work as expected. - All images display properly and as specified in the brief. - Appropriate use of folders to organize files on the web server. - Appropriate file names (based on best practices as described in brief). - Appropriate use of HTML and CSS comments.	
SD 1c Hosting a website LOs 16 Create a domain and host a website on it, students can use paid for, free or local hosting.	

SD 1d	Testing a website LO 11 LO 8
	• Test the completed website with regard to • Usability - effective navigation, clarity of presentation • Accessibility –ensure the site is accessible through screen readers, etc. • including, • Using alternate text in image tags • Using appropriate HTML5 tags, e.g. article, header, section, etc. • Cross browser compatibility
Skills Demonstration 2	15%
The learner will complete an assignment over a period of no more than 3 to 4 weeks	
Over multiple sessions throughout the year as the required learning outcomes have been delivered, the learner will demonstrate skills from learning outcomes LO 9 LO 14 The assessor will devise a brief which will require students to demonstrate: • Use of Client Side Scripting Use client side scripts to and include any six of the following: Validate form input, sliding panels, cycling images, image gallery, update totals, rollovers, open window, dynamic update, cookies. • Use of Server Side Scripting: Use Ajax to perform asynchronous updates to a page	
Skills Demonstration 3	10%
The learner will complete an assignment over a period of no more than 2 weeks	
Over multiple sessions throughout the year as the required learning outcomes have been delivered, the learner will demonstrate skills from learning outcomes LO 6 The assessor will devise a brief which will require students to create a website using CMS • Installing a CMS account i.e. WordPress, Joomla account • Adding Pages and Navigation in CMS i.e. WordPress, Joomla • Working with templates • Build content by creating new posts • Categorize and arrange post into an organised fashion • Categorize and sort pages into an organised fashion	

12. Grading

Distinction:	80% - 100%
Merit:	65% - 79%
Pass:	50% - 64%
Unsuccessful:	0% - 49%

At levels 4, 5 and 6 major and minor awards will be graded. The grade achieved for the major award will be determined by the grades achieved in the minor awards.

Web Development 6N1277	Learner Marking Sheet 1 Skills Demonstration 60%
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Learner's Name: _____

Assessment Criteria	Maximum Mark	Learner Mark
Skills Demonstration 1 30% - Design a website showing use of current standards in Html and CSS, appropriate use of colors and unit sizes - Create a website using appropriate use of URLs, links, images, folders, and filenames. Correct use of Html and CSS comments - Host a website showing an understanding of typographical and accessibility concepts - Test a Website	5 15 10 5	
Subtotal	35	
Skills Demonstration 2 20% - Create a webpage using CMS - Install account, create pages with navigation, use a template. - Style the page, add new posts.	5 6 4	
Subtotal	15	
Skills Demonstration 3 10% - Use of Client Side Scripting Use client side scripts to and include any six of the following: Validate form input, sliding panels, cycling images, image gallery, update totals, rollovers, open window, dynamic update, cookies. - Use of Server Side Scripting: Create a script to connect to a database server and retrieve some data - Write code that demonstrates common ways of displaying results from a database query, e.g. tables, lists	6 2 2	
Subtotal	10	
Total Mark	60	

Assessor's Signature: _____

Date: _____

External Authenticator's Signature: _____

Date: _____

Web Development 6N1277	Learner Marking Sheet 2 Examination Theory 40%
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Learner's Name: _____

Assessment Criteria	Maximum Mark	Learner Mark
Section A: Answer all questions		
Question No.1	8	
Question No.2	8	
Question No.3	8	
Question No.4	8	
Question No.5	8	
Total Mark	40	

Assessor's Signature: _____

Date: _____

External Authenticator's Signature: _____

Date: _____